4. Kalai wants to cut identical squares as big as she can, from a piece of paper measuring 168mm and by 196mm. What is the length of the side of the biggest square? (To find HCF using repeated subtraction method)

Objective Type Questions

5. What is the eleventh Fibonacci number?
   (a) 55 (b) 77 (c) 89 (d) 144

6. If F(n) is a Fibonacci number and n =8, which of the following is true?
   (a) F(8) = F(9)+F(10) (b) F(8) = F(7)+F(6) (c) F(8) = F(10)×F(9) (d) F(8) = F(7)–F(6)

7. Every 3rd number of the Fibonacci sequence is a multiple of _____
   (a) 2 (b) 3 (c) 5 (d) 8

8. Every _____ number of the Fibonacci sequence is a multiple of 8
   (a) 2nd (b) 4th (c) 6th (d) 8th

9. The difference between the 18th and 17th Fibonacci number is
   (a) 233 (b) 377 (c) 610 (d) 987

10. Common prime factors of 30 and 250 are
    (a) 2 x 5 (b) 3 x 5 (c) 2 x 3 x 5 (d) 5 x 5

11. Common prime factors of 36, 60 and 72 are
    (a) 2 x 2 (b) 2 x 3 (c) 3 x 3 (d) 3 x 2 x 2

12. Two numbers are said to be co-prime numbers if their HCF is
    (a) 2 (b) 3 (c) 0 (d) 1

7.7 Cryptology

In today's world, security in information is a fundamental necessity not only for military and political departments but also for private communication. Today's world of communication has increased the importance of financial data exchange, image processing, biometrics and e-commerce transaction which in turn has made data security an important issue. Cryptology is defined as the science which is concerned with communication in secured form.

7.7.1 Cryptology – Some technical details



Plain text: The original message is called plain text.

Cipher text or Cipher number: The encrypted output (converted message into code) is called Cipher text or Cipher number. Cipher text is written in capital letters, while plain text is usually written in lowercase. A secret key is to use something to generate the Cipher text from the plain text.

Encryption and Decryption: The process of converting the plain text to the Cipher text is called encryption and the vice versa is called decryption.

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